

HAT-P-36b

R-0891

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240220 · created 2026-07-29T02:08:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460360.8871 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1573 +/- 0.0064
Transit depth (Rp/Rs)^2	2.48 +/- 0.2 [%]
Orbital Inclination (inc)	89.66 +/- 1.66
Airmass coefficient 1 (a1)	1.022 +/- 0.014
Airmass coefficient 2 (a2)	-0.023 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#3 - [61, 61]
Optimal Aperture	3.75
Optimal Annulus	8.83
Transit Duration (day)	0.10015 +/- 0.00099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1573 ± 0.0064 vs 0.126 (deviation 4.09σ)
Duration fit vs published	0.10015 d vs 0.0925 d (deviation 6.47σ)
Mid-time O–C	1.6 min
Depth SNR	20.6
Residual scatter	1.36 %

Light curve

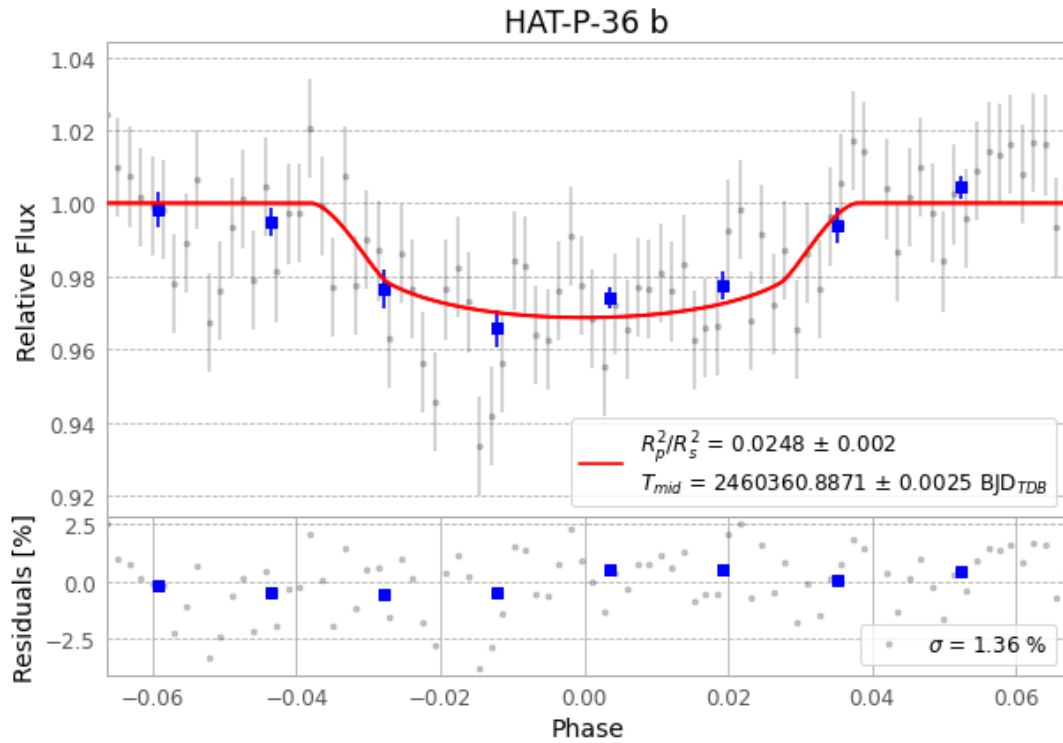


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-02-20.png (SHA-256 80ab14188123e486...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aacb61a021e261ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

86 FITS frames (56 MB), HATP-36240220070315.FITS → HATP-36240220111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240220.log (3f92502e9a64...), FinalLightCurve_HAT-P-36 b_2024-02-20.png (80ab14188123...), FinalLightCurve_HAT-P-36 b_2024-02-20.csv (29beeebf1f69...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:08Z.